

CERN detector could help to improve head tumour radiotherapy

Scientists are testing a new device to help target cancer cells more accurately in ion radiotherapy of head and neck tumours, which could help limit the treatment's side effects. It includes Timepix3, a small particle detector developed at CERN



Mária Martišíková (left), the project leader from Heidelberg University Hospital and German Cancer Research Center (DKFZ), and DKFZ researcher Laurent Kelleter. (Image: Heidelberg University Hospital / H.Schroeder).

Particle detectors like the ones used by physicists at CERN can have wide applications beyond fundamental research. Scientists from the German National Center for Tumor Diseases (NCT), the German Cancer Research Center (DKFZ), and the Heidelberg Ion Beam Therapy Center (HIT) at Heidelberg University Hospital are now testing a new imaging device supplied by the Czech company ADVACAM on its first patients. The device, which includes a small Timepix3 pixel detector developed at CERN, allows head and neck tumours to be closely monitored during ion radiotherapy, making them easier to target and thus helping limit the treatment's side effects.

"One of the most advanced methods for treating head and neck tumours involves irradiation with ion beams. This has one unique feature: it can be precisely tailored to the depth inside the human head where the particles should have the maximal effect", explains Mária Martišíková, the head of the DKFZ team. Yet like other types of irradiation, ion radiation also has a drawback. The particle beams affect not only the tumour but also part of the healthy tissue around it. This is particularly challenging in the brain, where damage to the optic nerve or a patient's memory are possible.

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Ideally, the irradiated area around the tumour should be as small as possible, and the dose to the tumour should be as high as possible. However, current technology does not allow for sufficiently precise targeting of the ions.

To complicate matters further, the situation inside a patient's head can change during therapy. The x-ray computed tomography (CT) scan image taken before treatment is essentially used as a "map" to target the tumour with ion beams. But during therapy, the situation inside the skull may evolve. Until now, physicians lacked a reliable tool to alert them in case of a change in the brain.

The new ADVACAM device could help solve these issues, by improving the navigation of the ion beams inside the head by tracking the secondary particles that are created when ions pass through it.

"Our cameras can register every charged particle of secondary radiation emitted from the patient's body. It's like watching balls scattered by a billiard shot. If the balls bounce as expected according to the CT image, we can be sure we are targeting correctly. Otherwise, it's clear that the 'map' no longer applies. Then it is necessary to replan the treatment," describes Lukáš Marek from ADVACAM.

"We hope the new device will show us how often and where the tumour changes occur. It will allow

us to reduce the overall irradiated volume of tissue, saving healthy tissue and reducing the side effects of radiotherapy. We will also be able to apply higher doses of radiation to the tumour" adds Martišíková.

The treatment can benefit enormously from the additional information obtained from the camera. In the first phase, data could lead to an interruption and replanning of the irradiation series when necessary. The ultimate goal is a system that can correct the path of the ion beam in real-time.

This device exemplifies successful knowledge transfer, showcasing how technology initially developed for detectors used in fundamental physics research can be applied in healthcare.

"When we started developing pixel detectors for the LHC we had one target in mind – to detect and image each particle interaction and thereby help physicists to unravel the secrets of Nature at high energies. The Timepix detectors were developed by the multidisciplinary Medipix Collaborations whose aims are to take the same technology to new fields. Many of those fields were completely unforeseen at the beginning and this application is a brilliant example of that," says Michael Campbell, Spokesperson of the Medipix Collaborations.

Accelerator Report: Beams are circulating in the LHC

On 8 March, three days ahead of schedule, the first proton beam was injected into the LHC; 20 minutes later, the second beam was injected, circulating in the opposite direction.

Since the last Accelerator Report, the hardware tests and subsequent cold check-out were successfully completed, both ahead of schedule. Once the usual remaining wrinkles were ironed out, everything was ready to start the 2024 LHC beam commissioning. The single bunch low intensity probe beam, meticulously prepared in the injector chain in the past weeks, came knocking at the LHC's door.

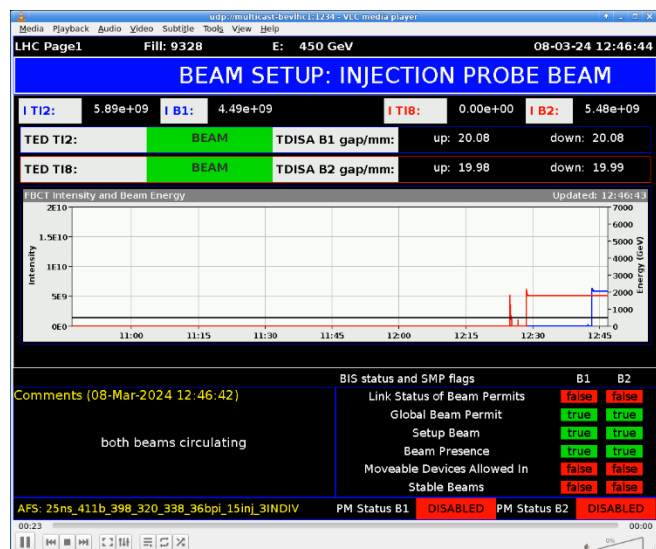
Many of the LHC engineers in charge and system experts gathered in the CERN Control Centre (CCC) on 8 March, alongside members of the Management, to witness the process, eagerly

waiting for the first beams to circulate again in the LHC.

The LHC Operations team started the injection and threading process for beam 2 (circulating counter clockwise): they injected the beam at LHC Point 8, just in front of the LHCb experiment, and let it circulate up to Point 7, where a set of collimators was fully closed to intercept it. The measurements performed by the beam position monitors indicated that the beam trajectory could be improved. This was quickly done using an automated beam steering tool that powers corrector magnets to smoothen the trajectories of the particles.

Confident in this correction, the Operations team opened up the collimators at Point 7 and closed the ones further along the ring at Point 6, before

injecting the beam again. This process was repeated until the last collimators, at Point 1 (ATLAS experiment), were opened, leaving the way clear for the beam to make a second, third, fourth... and millionth turn.



The LHC fixed display, just after both beams entered into circulation. The blue and red lines represent the number of protons in beams 1 and 2, respectively. The black line represents the energy of the beams. It is flat because the beams had not yet been accelerated at this point. (Image: CERN)

Another small correction to adjust the orbit of the circulating particles was made before attention switched to beam 1, which ended up circulating in the machine less than 20 minutes after beam 2 and was welcomed by many happy faces in the CCC. The next step – accelerating both beams up to 6.8 TeV – was also accomplished during the

weekend. Witnessing both beams in circulation is something of a relief for everyone involved, although the real beam commissioning work starts at that point.

For the 2024 run, it was decided to modify the optics of the accelerator and to replace them by reverse polarity optics (RP-optics). The objective is to mitigate the radiation suffered by some of the magnets of the inner triplet region on both sides of the ATLAS experiment. The inner triplet is a set of quadrupole magnets that focus the beam to very small dimensions at the centre of the experiments.

Some of the collision debris – particles produced by the collisions and travelling parallel to the beams, outside the experiment – is intercepted by the magnets in the inner triplet regions, inducing radiation damage to their insulation. With different optics, the debris is deposited in other places in these magnets, so that the burden of the radiation damage is distributed more widely. This helps to extend the magnets' lifetimes, even with an increased number of collisions.

The commissioning and validation of the RP-optics are among the many beam commissioning steps that have to be taken in the coming weeks before beams enter into collision at 6.8 TeV, hopefully on 8 April. Depending on how work progresses, this milestone may shift forwards or backwards by a few days.

Rende Steerenberg

CERN70: The dark side of the muon

Part 5 of the CERN70 feature series. Francis Farley, a British physicist, joined CERN in 1957. This marked the start of a long and remarkable career in experiments to measure the anomalous magnetic moment of the muon

In the 1950s, the muon was still a complete enigma. Physicists could not yet say with certainty whether it was simply a much heavier electron (with 200 times the mass) or whether it belonged to another species of particle. Acting on an idea of Leon Lederman, CERN launched the "g-2" experiment in 1959, aimed at measuring one of the properties of this strange electron – its magnetic moment.

The experiment's aim was to test quantum electrodynamics, a theory elaborated in the 1940s to describe the effect of the electromagnetic force on charged subatomic particles such as electrons or muons. Among other things, it predicts an anomalously high value for the muon's magnetic moment "g", hence the name of the experiment.



Installation of the g-2 experiment's muon storage ring at the end of the South hall, photographed in 1965. (Image: CERN)

A group of six physicists – Francis Farley, Georges Charpak, Théo Müller, Antonino Zichichi, Johannes Cornelius Sens and Richard Garwin – joined forces to try and measure this famous value on the Synchrocyclotron. In 1961, the team published the first direct measurement of the muon's anomalous magnetic moment to a precision of 2% with respect to the theoretical value. This precision was increased to 0.4% in 1962. The theory of quantum electrodynamics was validated; the muon behaved exactly like a heavy electron.

Francis Farley proposed a follow-up project at the Proton Synchrotron (PS), involving Simon Van der Meer and Emilio Picasso. Launched in 1966, the experiment produced results that were 25 times more accurate. A third experiment was launched in 1969. The final results confirmed the theory with a precision of 0.0007%! Research into the muon's anomalous magnetic moment continued in the United States, first at Brookhaven National Laboratory and then at Fermilab, where an experiment is still under way, with increasingly precise results.

Recollections

Francis Farley, a British physicist who joined CERN in 1957, put together the first "g-2" experiment on the anomalous magnetic moment of the muon. In the course of his long and remarkable career, he conducted a number of experiments to measure this value with ever greater precision.

"One day I received a telegram from the magazine Scientific American asking me to send details of my experiment for an article about the Moon. As my experiment had nothing to do with the Moon, I didn't reply. A few weeks later, however, they wrote to me again. It was then that I realised that

there had been a typing mistake. They were actually asking me for information about the muon, not the Moon. At first, I had thought that there was no connection between the two, but I was wrong.



Team members of the “g-2” experiment at the Synchrocyclotron (SC) in 1961. Francis Farley stands next to (left to right) Johannes Cornelius Sens, Georges Charpak, Théo Müller and Antonino Zichichi, sitting on the experiment's 6-metre-long magnet. (Image: CERN)

One of the great mysteries at the time of Isaac Newton was understanding how the Moon was affected by the Earth or, in other words, how gravity worked. It's the same thing with electromagnetics. What happens between two charged particles that are attracted to each other or repel each other? This is what the theory of quantum electrodynamics explains. It holds that the electron is surrounded by a cloud of “virtual” photons that are emitted and reabsorbed. If another electron enters the cloud, photons are exchanged.

Gravitation is explained by a similar theory. Virtual gravitons are thought to be exchanged between the Sun, the Earth and the Moon. In quantum

electrodynamics, the virtual photons alter the magnetic moment of the particle. If we can detect this variation, we have direct evidence that these virtual photons exist and this was the aim of our experiment.

The first article that we published, in 1961, took the world by storm, because it was the first time that the anomalous magnetic moment of the muon had been measured. As a result, the experiment served as a launching pad for group members to go on to do other things. I was the only one who stayed behind. I went on to work on the second experiment, where I met Emilio Picasso, with whom I got on tremendously well. We got a new team together and achieved much more precise measurements than with the first experiment. As our results did not tie in with theory, we started to think of ideas for a third experiment. We achieved such precision that we decided not to go any further.

The science I have experienced has been all about imagining and creating pioneering devices and observing entirely new phenomena, some of which have possibly never even been predicted by theory. That's what invention is all about and it's something quite extraordinary. CERN was marvellous for two reasons: it gave young people like me the opportunity to forge ahead in a new field and the chance to develop in an international environment.”

This interview is adapted from the 2004 book “Infinitely CERN”, published to celebrate CERN's 50th anniversary. Francis Farley [passed away in 2018](#), aged 97. Read more about the history of g-2 in the [CERN Courier](#).

The Open Quantum Institute launches its pilot phase at CERN

The three-year pilot will build on the efforts to date to help unleash the full power of quantum computing for the benefit of all

The [Open Quantum Institute](#) (OQI) passes a new milestone today, with the operational launch at CERN. Following a successful one-year incubation period led by the [Geneva Science and Diplomacy Anticipator](#) (GESDA), the new, three-year CERN-

based pilot will build on the efforts to date to help unleash the full power of quantum computing for the benefit of all.

Proposed, designed, and incubated through GESDA, in collaboration with some 180 experts

from all over the world, the OQI is a multilateral science diplomacy initiative, uniting academia, technology companies, the private sector, the diplomatic community, philanthropy organisations and global citizens in a joint effort towards more open and inclusive quantum computing. By facilitating equal access to cutting-edge nascent technologies, the OQI seeks to accelerate the potential of quantum computing for all society and to support the development of concrete quantum solutions aimed at achieving the United Nations' Sustainable Development Goals (SDGs).

During its pilot phase, hosted at CERN and supported by the Union de Banques Suisses (UBS), the OQI will be part of CERN's wider [Quantum Technology Initiative](#) (QTI), launched in 2020 and managed by the IT department. Building on QTI's mission to explore the full potential of quantum technologies and to maximise their societal impact, the OQI will work to push the boundaries of geography and disciplines to ensure that quantum computing is harnessed to tackle some of the key global challenges.

"CERN offers ideal conditions for the development of the OQI, and my hope is that this initiative will not only be a success, but also a model of what scientific diplomacy can do to promote concrete projects of benefit to humanity", says Fabiola Gianotti, CERN Director-General. "During the pilot phase, the OQI will benefit from CERN's experience in deploying scientific and technological progress to the benefit of society. We look forward to working with GESDA and other partners from academia, industry and government to ensure that quantum computing is accessible to all, including underserved regions of the world."

The focus will lie on the selection of SDG-related use cases to explore applications of quantum computing in fields like health, energy, climate action, clean water, and food security. Some

examples of potential projects include: improving the sustainability of global food systems through quantum computing optimisation (addressing SDG 2, zero hunger); finding quantum machine learning solutions to improve medical imaging accuracy and early diagnosis of diseases (addressing SDG 3, good health and well-being); and using quantum computing simulation to reduce carbon dioxide in the atmosphere (addressing SDG 13, climate action).

"The UN's SDGs represent the international community's collective view of what the greatest societal challenges are today," says Enrica Porcari, Head of CERN's IT department. "This is why we are proud to host the OQI at CERN and to provide a platform for innovation, fostering real-world applications of quantum computing to address the SDGs."

CERN will host the OQI from 2024 to 2026 and support three or four projects targeting SDG-related use cases. It will also lay the foundation for the next phase of the programme and potentially become a reference point for other initiatives aimed at deploying quantum technologies to address societal challenges. GESDA will remain the science diplomacy advisor and fundraiser, helping to ensure the continuity of the initiative and contributing to its diplomatic engagement, while UBS will act as the lead support partner, ensuring further growth of the institute.

Organisations and individuals, committed to human-centred, inclusive and responsible quantum computing, can play their part in OQI by submitting use cases for SDGs, developing educational tools, curating the diplomatic dialogue on quantum computing and much more. *For full details on how to get involved, please visit the [website](#), and follow OQI on [LinkedIn](#) and [X](#).*

Anastasiia Lazuka

You see an empty field? We see an "Open Sky Laboratory"!

Solutions proposed in the Future Circular Collider "Mining the Future" competition will now be evaluated in a 10 000 square metre "Open Sky Laboratory"

The Future Circular Collider ([FCC](#)) Feasibility Study is developing a concept for a new research

infrastructure to host the next generation of higher-performance particle colliders with the aim

of extending the research currently being conducted at the LHC, once the HL-LHC reaches its conclusion, beyond 2040.

In 2021–2022, the EU-funded FCC Innovation Study* launched an international challenge-based competition called “[Mining the Future](#)”, which invited scientists and companies to propose innovative yet technically feasible solutions to turn the material excavated during the construction of FCC underground structures into a usable resource. The reuse potential of the excavation material is one of the factors that will contribute to the acceptability and cost efficiency of the FCC project.

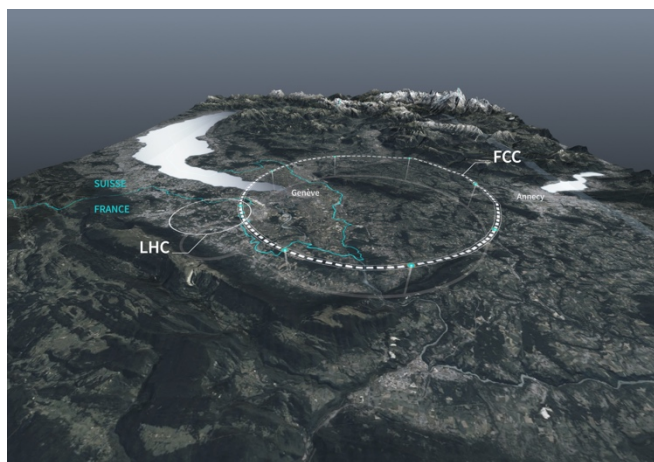


Diagram illustrating a potential placement scenario for the FCC underground tunnel. (Image: CERN)

The proposed solutions are now being integrated into a unique design and evaluated in the field, and will reach maturity by 2030. The objectives of the evaluations are twofold. Firstly, to establish how to conduct the online identification, sorting and pre-treatment of the materials during the excavation process. Secondly, to prepare different reuse pathways to sort and pre-treat materials, including transforming sterile rock – a soft and heterogeneous sedimentary rock called molasse –

into fertile soil for agriculture, forestry and renaturation applications, in line with the principles of a circular economy. The quality-assured creation of fertile soil is a lengthy process spanning several years and has been chosen as the first large-scale experiment with field tests at an “Open Sky Laboratory”.

The Open Sky Laboratory, a plot of about 10 000 m² located near LHC Point 5 (CMS, Cessy, France – see image), has been made available and will be prepared in collaboration with CERN’s SCE and EN departments. Molasse extracted during the HL-LHC excavations will be transported to this field to be used in the tests. Initial laboratory analyses will be performed off site to identify the most suitable mix of molasse and other materials. These will be followed by field tests in the Open Sky Laboratory’s controlled environment (monitoring of the field, weather and plant growth conditions), using scientific protocols developed by a collaboration of universities working in this domain.

In keeping with CERN’s long-standing tradition, this project relies on an open collaboration with academia and industry. Currently, the collaboration includes university and research experts in agronomy, pedogenesis and geology and industrial partners in soil engineering and phytoremediation, soil treatment techniques and monitoring and supervisory control systems.

A socioeconomic assessment of reuse cases for the transformed soil will be performed in order to evaluate the potential economic benefits for FCC construction and the potential advantages for the entire construction sector in Europe.

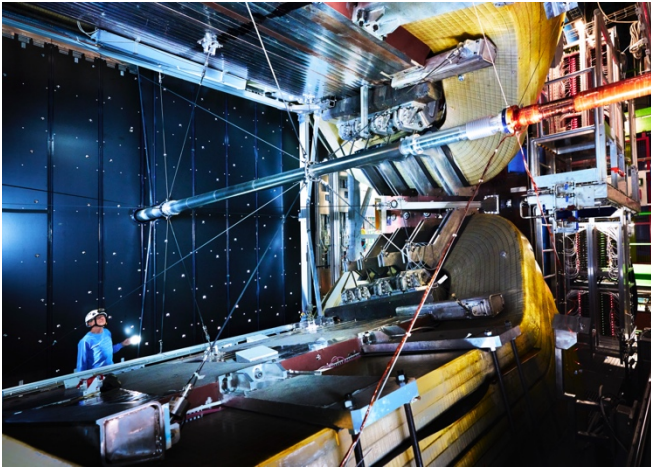
**Grant agreement 951754*

Luisa Ulrici

LHCb observes a new decay mode of the charmed beauty meson

The result has implications for future searches for rare beauty meson decays and for the interpretation of results from the Fermilab g-2 experiment

The LHCb collaboration recently reported the first observation of the decay of the B_c^+ meson (composed of two heavy quarks, b and c) into a J/ψ charm-anticharm quark bound state and a pair of pions, $\pi^+\pi^0$. The decay process shows a contribution from an intermediate particle, a ρ^+ meson that forms for a brief moment and then decays into the $\pi^+\pi^0$ pair.



The LHCb detector. (Image: Maximilien Brice, CERN)

The B_c^+ is the heaviest meson that can only decay through the weak interactions, via the decay of one heavy constituent quark. B_c^+ decays into an odd number of light hadrons and a J/ψ (or other charm-anticharm quark bound states, called “charmonia”) have been studied intensively and have been found to be in remarkable agreement with the theoretical expectations. The decay of B_c^+ into a J/ψ and a $\pi^+\pi^0$ pair is the simplest decay into charmonium and an even number of light hadrons. It has never been observed before, mainly because the precise reconstruction of the low-energy π^0 meson through its decay into a pair of photons is very challenging in an LHC proton-proton collision environment.

A precise measurement of the $B_c^+ \rightarrow J/\psi \pi^+ \pi^0$ decay will allow better understanding of its possible contribution as a background source for the study of other decays of B_c mesons as well as rare decays of B^0 mesons. From the theoretical point of view, decays of B_c into J/ψ and an even number of pions are closely related to the decays of the τ lepton into an even number of pions, and to the e^+e^- annihilation into an even number of pions. Precise measurements of e^+e^- annihilation into two pions in the ρ mass region (as in the B_c decay discussed here) are crucial for the interpretation of results from the Fermilab g-2 experiment measuring the anomalous magnetic dipole moment of the muon, since low-energy e^+e^- annihilation into hadrons is an important source of the uncertainty of the g-2 measurements.

The ratio of the probability of the new decay to that of the decay of B_c^+ into $J/\psi \pi^+$ has been calculated by various theorists over the last 30 years. Now these predictions can finally be compared with an experimental measurement: most predictions agree with the new result obtained by LHCb ($2.80 \pm 0.15 \pm 0.11 \pm 0.16$).

The large number of b-quarks produced in LHC collisions and the excellent detector allows LHCb to study the production, decays and other properties of the B_c^+ meson in detail. Since the meson’s discovery by the CDF experiment at the Tevatron collider, 18 new B_c^+ decays have been observed (with more than five standard deviations), all of them by LHCb.

Read more in the LHCb paper: <https://cern.ch/5hsse>

LHCb collaboration

Reducing emissions related to duty travel: everybody's contribution counts

New recommendations provide guidance for making environmentally responsible travel choices

In an era where the consequences of climate change loom ever larger, reducing carbon emissions is imperative. Among the various contributors to carbon emissions, duty travel – comprising conferences and other professional engagements – is an often-overlooked source.

Travel is embedded in CERN's DNA, with worldwide collaborations that rely on exchanges between people from all around the globe. International exchange is a pillar of scientific progress at CERN and beyond, and is particularly important for early-career researchers and those from underrepresented geographic regions.

At the same time, CERN strives to be a role model for environmentally responsible research and, in this context, to minimise its carbon emissions wherever possible.

Extensive professional travel, particularly by air, has an environmental impact. [CERN's third Environment Report](#) shows that emissions arising from travel by personnel on the CERN payroll amounted to 151 tCO₂e and 827 tCO₂e in 2021 and 2022 respectively, a marked reduction compared with the 3330 tCO₂e reported in 2019, before the COVID-19 pandemic. Most emissions result from air travel, mainly from long-distance flights. Although they represent just a small fraction of CERN's total emissions, acting on them is important as, whatever the scale, all actions contribute to minimising CERN's environmental impact.

A dedicated working group was set up in 2022 to make recommendations on reducing duty-travel emissions without having a detrimental impact on CERN. The recommendations, approved by the Enlarged Directorate on 23 January 2024, recognise and integrate the crucial importance of international collaboration for the advancement of CERN's mission and research, while encouraging

everyone to collectively set an example by reducing duty-travel-related carbon emissions.

The recommendations are now available in the [Admin e-guide duty travel pages](#).

Two overarching principles apply:

1. Reducing (air) travel by considering whether virtual participation, if available, provides similar value. This will help reduce the number of trips undertaken and contribute to reducing overall emissions. **In particular, single-day trips requiring air travel should be avoided if virtual participation is possible.**
2. Encouraging the use of ground transportation (particularly the train) for distances up to 700 km and as transport options allow, taking into account time- and cost-efficiency.

The recommendations are accompanied by a simple decision tree to help travellers reflect before they book their trips. In addition, the new online booking tool [Egencia](#) offers several different features, including a CO₂ calculator for flights. Finally, the recommendations also cover [event guidelines](#) to encourage organisers and participants to make mindful, environmentally conscious choices. These take into account the importance of in-person interactions for building and maintaining collaborations and networks, particularly for early-career professionals. They aim to ensure that effective virtual participation is possible in all events hosted by the Organization and to minimise the amount of travel required without compromising collaboration, operations, goals and opportunities for personnel.

HSE Unit

CERN community: celebrate spring at CERN with us



Send us your “spring at CERN” photos. (Image: Naomi Dinmore/CERN)

Blossom on the trees, longer days and a strange craving for egg-shaped chocolates tells us that spring is in the air.

To celebrate, we’re launching a photography competition for the CERN community. Send us your best photos of “spring at CERN”. Our favourite photo will win a CAGI Chocopass, kindly offered by the CAGI cultural kiosk at CERN. This Chocopass lets you spend a day exploring Geneva and tasting from a range of chocolate shops.

How to enter:

- You must have a CERN email address to take part in this competition.
- Send your photo to bulletin-editors@cern.ch by Sunday 24 March at 11:59 p.m. CET.
- You can send a maximum of three photos per person.
- By sending your photo, you agree to it being added to a CC-BY photo collection in the CERN Document Server, where you will be credited.
- The photo may be used by CERN online for this competition and in the future.
- The winner will be announced in the next CERN Bulletin.

We are grateful to the International Geneva Welcome Centre (CAGI) for offering a Chocopass for the winning prize. The CAGI cultural kiosk is located in CERN’s main building and is open from Monday to Friday from 8:30 a.m. to 11:00 a.m. and from 11:30 a.m. to 2:30 p.m. It offers numerous discounts for local activities and events both in Switzerland and in France. Find out more here: <https://www.cagi.ch/en/cultural-kiosk-agenda/>

Internal Communication

Enabling open access to books

An expanded collaboration agreement between CERN and the OAPEN (Open Access Publishing in European Networks) Foundation sees the Laboratory directly hosting the OAPEN Library and the Directory of Open Access Books in its Data Centre

CERN and the not-for-profit organization [OAPEN Foundation](#) are happy to announce a further expansion of their collaboration to jointly promote open access to books.

Since 2021, CERN and the OAPEN Foundation have collaborated to disseminate books that are made available in open access through [SCOAP³ for Books](#), a collective open-access initiative hosted at CERN.

Building on this successful track record and the aligned values and goals, OAPEN and CERN have

now signed an expanded collaboration agreement whereby, from 2024 onwards, CERN will use its extensive technical infrastructure to directly host both the [OAPEN Library](#) and the [Directory of Open Access Books \(DOAB\)](#) in its [Data Centre](#), alongside its other scholarly communication services such as [INSPIREhep](#) and [Zenodo](#).

This mutually beneficial arrangement will help increase the operational efficiency and reliability of OAPEN and DOAB as vital services for the research community spanning all scholarly

disciplines. “For CERN, this further amplifies its existing efforts to support and promote open science,” says Alexander Kohls from the [CERN Scientific Information Service](#). “Utilising the CERN infrastructure will significantly increase stability for researchers worldwide allowing free and open access to the more than 30,000 books hosted in the OAPEN Library. Moreover, leveraging this infrastructure across multiple initiatives and disciplines ensures efficiency.”

Since the very beginning of both organisations, open research has been at the core of their

respective missions. CERN has already successfully supported other communities in establishing or expanding open science principles and collaborates with organisations such as UNESCO, the European Commission, and NASA to further accelerate the global transition to open science. For more than a decade, OAPEN has operated three platforms dedicated to the dissemination and discovery of open-access, peer-reviewed books.

Farewell to the Alcatel phone exchange

On 14 February, CERN disconnected the Alcatel phone exchange, which began in 1990 (and hit the [frontpage of the Bulletin](#) at the time). This now marks the end of physical office telephones in the Laboratory, with the migration to the software-based systems – TONE and CERNphone – now completed.



Disconnecting the Alcatel phone exchange on 14 February: (left to right) IT Department Head Enrica Porcari, Tony Cass, Rodrigo Sierra and Ihor Olkhovskiy. (Image: CERN)

“We replaced the Alcatel phone exchange, which routes calls within CERN and to outside lines, with a CERN-built exchange called TONE,” explains Tony Cass, head of the IT Communications Systems group. Last year, the TONE exchange also took over the features of the old “call centre” system that handles calls to, e.g., the Service Desk where several people respond to calls to a single number.

“The harder task,” he continues “was to build the software phone application and replace all the physical phones by software phones.” After considering various alternatives and providers, rollout of two CERNPhone applications started at the end of 2020. Read more details in the [IT README](#).

When CERN began in 1954, “the CERN phone directory contained everyone’s home number” as Franco Bonaudi [recalls](#) in our [CERN70](#) series. By [1965](#), just 17 phone lines served 1000 CERN extensions, with callers facing long waits. Now, as well as TONE and [CERNphone](#) for phonecalls, CERN also offers the [CERN Campus App](#) for on-site information, news and events.

Computer Security: Bingo walk-through

Once more, bravo to all those who participated in the Bull**** Bingo in the last but one Bulletin issue and sent us their solution. The Hawaiian pizza was quickly gone. But some people were wondering why their responses were not correct... Good point, so let's walk through that Bingo:

- 1A "There is no malware for Apple devices" – was a slogan of the past, as the big market share was with Microsoft Windows. But that has changed. And owners of a MacBook are perceived as being richer, so the spoils are larger...;
- 1B "Software from the Google Play Store is harmless" – actually, anyone can push any application to the Google Play Store. It is not curated or validated so it has become a dump for malware, too. The Apple store is better in that respect, as Apple keeps it tightly controlled (to make money) and only admits applications that follow their policies (which doesn't mean that [there are no problems either](#));
- 1C "Security is everyone's responsibility" – indeed! Given the academic environment at CERN, its openness and the freedom you have to benefit from any (legal) computing resource, it is impossible for the Computer Security team to shoulder the responsibility for your digital assets. We have to count on you to keep those secure, and we are happy to help you do so;
- 1D "SSH on port 2222/tcp is more secure" – nope. This is called security-through-obscurity as SSH normally runs on port 22/tcp. Changing that might only deter the lightweight attackers and, thus, keep the noise level down. The experts run reconnaissance tools like "[nmap](#)", which find SSH regardless of which port you use;
- 1E "Spam and malware filtering is 100% effective" – we wish! But in CERN's environment, with the openness to use the CERN email address for personal matters, with the multitude of languages spoken and written at CERN, and with the large academic community sending emails back and forth, for many emails it is hard to tell

[whether they are spam or not](#). We try our best, but, admittedly, there is room for improvement;

- 2A "2FA is a big step forward for account protection" – it is! With [two-factor authentication](#) it is no longer enough to remotely steal your password (e.g. via a so-called phishing attack). The attacker would also need to have physical access to your smartphone or hardware token, and most attackers are not close by. In addition, you would quickly notice the theft of your smartphone or keys, no?
- 2B "Emails from "@cern.ch" are legitimate" – [emails can be easily spoofed](#). That means that an email is not necessarily sent from a "cern.ch" mail service but from another one (e.g. gmail.com). This is called "spoofing" and is the reason why so many emails are currently quarantined in our mail appliances, because the sender is consciously or accidentally spoofing email addresses;
- 2C "I'm personally not a target as I'm not interesting to attackers" – don't be so humble. You are interesting (!) even if you might not be the one working on that very confidential, top-secret or highly visible stuff. You might just be the entry point. The personal assistant used to attack the boss. The colleague to trick your peers. The patient zero to infect and compromise others...
- 2D "Back-ups cannot be altered" – as long as they remain connected and remotely reachable and are not immutable, most back-ups can be altered. [CERN takes special care](#) to ensure that back-ups are secure, but if you leave your external hard-disk connected to your laptop, it is for some malware only another folder to subject to [ransomware encryption](#)...
- 2E "I have nothing to hide" – don't you? Can I get your credit card PIN? Install a camera in your place? Access the "deleted photos" folder on your smartphone?

- 3A “I would never fall for phishing” – said many other people before. We usually catch out 10% of CERN accountholders with our [annual phishing campaigns](#);
- 3B “Only the link behind a text/QR code reveals its truth” – yessss! What is displayed can be anything (for readability or for obfuscation). Only once you [hover your mouse over the link](#) or [check the text displayed right before taking the QR photo](#), is the real destination revealed.
- 3C “CERN’s technical network is secure” – it is secured. But given its complexity, its automatic interaction with CERN’s Data Centres and the need for experts and operators to remotely connect to the technical network, [it is far from being perfectly secure](#). There is still some margin for improvement!
- 3D “A password written on a post-it is a good idea” – if you want to give the cleaning personnel or visitors access to your computer. And you shouldn’t!
- 3E “QR codes always link to legit sites” – nope. There is no guarantee of that (see 3B above);
- 4A “A (free) VPN service protects me” – but remember, if you’re not paying for the product, there’s a very high chance that you are the product. There is no guarantee regarding anonymity or privacy with a free VPN, and the provider is free to share your data with third parties as they see fit or to (ab)use your network bandwidth for other purposes. This is why, for example [“Hola! VPN” is forbidden at CERN](#). Paid VPNs are better, but still protect only your communication, they do not protect against the content you access. In the end, it is a question of whom you trust more, your ISP (and thus indirectly the country/jurisdiction under which that ISP operates) or your VPN provider (and the country where they/their servers are located);
- 4B “Password protection on my laptop protects its data” – actually, that password protects interactive access to your laptop. But if you don’t take extra precautions to encrypt your hard disk with Bitlocker (Windows) or Filevault (MacOS), your laptop is for an attacker just another unprotected storage system like a USB stick;
- 4C “My browser’s password manager is secure” – that depends very much on the kind and on which version you run. In the past, [passwords were even stored in plain text in some browsers](#). Hence, if you can’t remember your passwords, a [stand-alone password manager might be the better choice](#);
- 4D “CERN is not interesting to attackers” – was never a true statement. The [Chaos Computer Club infiltrated CERN in 1986](#); at the beginning of the millennium we fought “Phalanx”, “Windigo” and “ebury” in our data centres and on the WLCG; and today [CERN is targeted by ransomware attacks like anyone else](#) ;
- 4E “CERN’s anti-malware software is free for you to download” – yes, it is! [For the protection of your devices at CERN and at home](#), and for the wider protection of CERN;
- 5A “Using “https” means the website is secure” – the “s” in “https” indicates that the communication is protected by encryption and, hence, from eavesdropping. But this doesn’t imply the trustworthiness of the website behind;
- 5B “CERN’s outer perimeter firewall keeps all threats away” – if that were true, we wouldn’t have this article here. While that firewall blocks a large fraction of malicious and unwanted traffic, it is not watertight and requires other [“defence-in-depth”](#) layers to catch all threats;
- 5C “Cloud services cannot be hacked” – actually, cloud service providers are in the same boat as everyone else and, in addition, are big targets with lots of revenue to be made. Indeed, there have been reports in the past of a multitude of successful attacks on large cloud service providers like [Okta](#), [Microsoft](#), [LastPass](#), etc.;
- 5D “Encryption is easy; key management is complicated” – true! There are a multitude of good encryption mechanisms on the

market. It gets tricky, however, to ensure that the decryption keys are properly and securely stored. If they get lost, so is your encrypted data. And it gets even more tricky if several people, each with their own decryption key, need to access the data;

- 5E “WiFi is always secure” – WiFi is just a communication method, here via the air. It does not say anything about access protection or encrypted communication. If you want a secure WiFi, ensure that the communication is subject to “WPA3” and, even better, always use encrypted protocols: SSH or HTTPS (see 5A above).

Complicated? Maybe. But that’s why we keep you regularly updated in our Bulletin articles. So, once

again, CONGRATULATIONS to those who got five right answers. And THANK YOU to you all for helping to keep CERN secure!

CERN Computer Security team

Want to learn more about computer security incidents and issues at CERN? Read our monthly reports

(https://cern.ch/security/reports/en/monthly_reports.shtml). For more information, questions or advice, check out our website (<https://cern.ch/Computer.Security>) or contact us at Computer.Security@cern.ch.

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

11–19 Mar | Computing | CERN | CERN IT Storage and Data Management group | EN

[TechWeekStorageCERN24](#)

15 Mar 11:00 | Knowledge Sharing | Online | CERN Alumni | EN

[Virtual Company Showroom with Sirona Technologies](#)

20 Mar 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN

[Status and Prospects of the JUNO neutrino experiment](#)

25–27 Mar | Computing | CERN | CERN openlab | EN

[2024 CERN openlab Technical Workshop](#)

18 Apr 19:30 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN

[CERN70 public event: The virtuous circle of knowledge and innovation](#) (more [CERN70](#) events)

Registration now open

15 May | Knowledge Sharing | Lausanne | Famelab | EN, FR or DE

[Famelab Switzerland local heats](#), open to 18–35-year-olds

23 May 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 10 – Artificial Intelligence \(AI\) & The Future of Work](#)

25–27 May | Accelerators | CERN | PBC | EN

[Physics Beyond Colliders Annual Workshop](#)

2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN

[Mechanical & Materials Engineering for Particle Accelerators and Detectors](#)

10–12 Jun | Knowledge Sharing | Online | Sustainable HEP | EN

[Sustainable HEP 2024 – 3rd Edition](#)

11–13 Jun | Engineering | CERN | FDF | EN
[1st FPGA \(Field Programmable Gate Arrays\) Developers' Forum meeting](#)

12–25 Jun | Physics | Nakhon Pathom (TH) | Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP) | EN
[2024 Asia-Europe-Pacific School of High-Energy Physics](#)

23 Jul–1 Aug | Knowledge Sharing | European Scientific Institute (FR) | I.FAST | EN
[Accelerators for healthcare? 10-day innovation challenge](#)

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN
[Introduction to Accelerator Physics](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Next blood donation campaign on 26 March

The poster is for a blood donation campaign at CERN. It features a red background with white and yellow text. At the top, it says 'BLOOD DONATION AT CERN' and 'IT'S IN YOUR BLOOD TO SAVE LIVES'. Below that, it states 'Donate your blood and receive a 10 CHF voucher valid in all CERN NOVAE restaurants.' The main text reads 'TUESDAY 26 MARCH 2024 FROM 8:30 TO 16:30 CERN, Building 29'. It also includes the website 'hse.cern/blooddonation' and a note to 'Save time, pre-book your appointment with OneDoc. Spontaneous donations also welcome.' There is an illustration of a person donating blood and the HUG logo at the bottom right.

At the last blood donation campaign on 28 November 2023, 121 donors came along to give blood, of which 40 were first-timers. 101 donations were collected.

The Geneva [blood barometer](#), which is regularly updated, shows that blood stocks continue to run extremely low. CERN will hold a third blood donation session on Tuesday 26 March, from 8.30 a.m. to 4.30 p.m. in collaboration with the Hôpitaux universitaires de Genève ([HUG](#)), in [Building 29](#).

Before coming to the Building 29 donation point, please make sure that:

- you are in good health and have no symptoms such as fever, cough, a cold or breathing difficulties;
- you are eligible to give blood;
- Complete the brief online [pre-donation questionnaire](#) (in French)

- Please consult this [information sheet](#) and complete this [questionnaire](#) issued by the HUG (but note that only the pre-donation conversation with the nurse or doctor on the day can confirm your eligibility).
- If returning from travel abroad or arriving from another country, perform the [travelcheck](#) on the HUG website (in French)
- If you have recently been vaccinated against the flu or COVID, ensure 48 hours have passed since receiving the vaccination.
- Please bring proof of ID with you on the day of the donation.

Save yourself time: pre-book your appointment with OneDoc

The possibility of pre-booking your appointment via the OneDoc platform is made available on this link <https://cern.ch/qzley>. Spontaneous donors will be able to come at their convenience without an appointment, mindful that there may be some waiting time on the day.

As a thank you, the HUG will be giving each donor a 10 CHF voucher, to be used in Novae's Restaurants 1 and 2 at CERN.

We hope to see many of you at the donation point!
[Give blood, save lives.](#)

HSE Unit

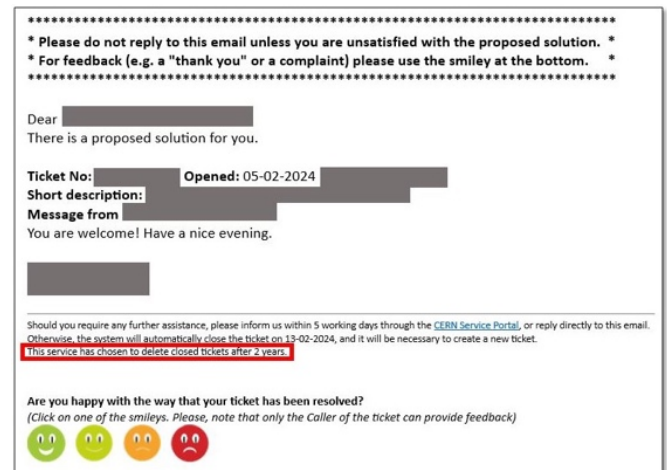
CERN ticketing system introduces automated retention – closed tickets may be deleted

With nearly 400 services across CERN making use of the [CERN Service Desk](#), you've probably had to open a few tickets over the years, from requesting a light bulb change to resetting your account password or safety questions. In fact, the Service Desk covers most of the things that a small town like CERN needs.

Those services and the tool used to keep track of your tickets were in place long before [Operational Circular OC11](#) on "The Processing of Personal Data at CERN" was introduced, and the Service Management teams have recently been working hard to strengthen the protection of your personal data in tickets in order to comply with OC11.

A recently introduced feature now allows each service to define how long your closed tickets will remain in the system. Services can choose from five different options to cover the various requirements across CERN: from five days to seven years and even indefinitely for certain cases, depending on the service's need to process and retain information. The [Privacy Notice](#) of each service will be updated accordingly.

When you receive a message that your ticket has been resolved, it will soon include information on when the ticket will be deleted from the system, as decided by that service:



This means that, when that date has been reached, the ticket will no longer appear in the "[My Items](#)" section of the Service Portal or in the ServiceNow tool.

The list of options and further details are available in [KB0009258](#). For more information on Data Retention, please refer to the [Guidelines](#) provided in the Admin e-guide.

IT department